OLAOLU ADENIYI

Senior AI / ML Engineer · API & Agentic System Development · Banking & FinTech

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Location: Lagos, Nigeria · Open to UK / US / Remote

PROFESSIONAL SUMMARY

Senior AI/ML Engineer and API developer with 5+ years designing and shipping production-grade intelligent systems in banking and financial services. Specialised in LLM-powered agentic architectures, conversational AI, NL2DAX/NL2SQL, and predictive analytics — with hands-on software development experience building REST APIs, WebSocket services, and serverless backends that bring AI models to production. Deep expertise across the full ML lifecycle — research, feature engineering, MLOps, deployment, and measurable business impact. Proven track record of systems serving millions of customers and generating billions in revenue uplift. Fluent across the Azure AI ecosystem, Microsoft Fabric, Python, and modern LLM orchestration frameworks. Currently pursuing MSc Data Science at the University of East London with research focused on AI applications in SEND.

CORE COMPETENCIES

AI & ML Engineering:
LLM Application Development · Agentic AI & MCP · RAG Architecture · Prompt Engineering · NL2DAX / NL2SQL · Conversational AI · MLflow · AutoML · Deep Learning · Predictive Modelling · Customer Segmentation · CLV Modelling · Recommendation Systems · Sentiment Analysis

API & Software Development:
REST API Design · FastAPI · Flask · WebSocket · Microservices · Docker · OpenAPI / Swagger · Serverless (Azure Functions) · Async Python · Event-Driven Architecture

Cloud & Azure:
Azure OpenAI · Azure AI Foundry · Azure Functions · Azure Data Factory · Microsoft Fabric · Power BI · Azure Cognitive Search · CosmosDB · Azure Table Storage · Azure Key Vault · Copilot Studio · AWS · Vertex AI

Languages & Frameworks:
Python · SQL · DAX · JavaScript · LangChain / LangGraph · scikit-learn · Keras · Pandas · Matplotlib · Power Automate · Git

WORK EXPERIENCE

Wema Bank Plc · Machine Learning Engineer *Mar 2024 – Present Lagos, Nigeria*

- **Databot — Conversational AI Agent on Power BI (NL2DAX)**

- Architected and deployed an enterprise conversational AI agent embedded in Databox (Wema Bank's internal Power BI platform), enabling 500+ relationship managers and branch staff to query live semantic models through plain natural language — eliminating the need for SQL or DAX expertise.
- Built the full production stack: Azure Functions (Python 3.11) for serverless compute, Azure OpenAI GPT-4o for DAX generation, Microsoft Fabric for semantic models, and Azure Table Storage for registry, audit logging, and conversation memory.
- Engineered a 29-rule DAX generation framework covering metrics table detection, customer vs account disambiguation, follow-up conversation context inheritance, Nigerian English dialect tolerance, PII-safe schema intelligence with cardinality-based value sampling, and ISBLANK() contact field detection.
- Deployed across 8+ dashboards covering ATC performance, VRM activities, SME loan tracking, AFB onboarding, foreign transactions, and market analytics — all from a single universal codebase with zero per-dashboard hardcoding.

✓ 500+ bank staff served ✓ 8+ dashboards live ✓ Sub-5s query response ✓ Zero-code analytics access

- **DataPilot — Autonomous Data Engineering Agent for Azure Data Factory**

- Designed and deployed an autonomous conversational agent enabling engineers to create end-to-end ADF pipelines, monitor health, diagnose failures, and trigger runs entirely through natural language in Microsoft Teams.
- Architected on Microsoft Copilot Studio backed by 15 Python Azure Functions on Azure Flex Consumption, exposed through a custom OpenAPI/Swagger 2.0 Power Platform connector.
- Implemented SOURCE/SINK parameter separation, automatic storage-type detection, async parallel execution via asyncio.gather, and a three-tier fuzzy schema mapper (confirmed, suggested, unresolved).

✓ Pipeline creation: Portal hours → <2 min conversation ✓ 15 microservices · 1 connector ✓ Junior engineers execute senior-level ADF ops

- **Risk AI Agent — Intelligent Credit Risk Assistant**

- Designed AI assistant automating review of bank statements, financial statements, and CAMs across Enterprise Risk Department and 3 pilot branches covering 3,000 accounts.
- Initial Copilot Studio architecture evolved to full Azure AI Foundry RAG stack: Azure AI Search, CosmosDB, Key Vault, containerised microservices, Zero Trust security — delivered with Microsoft Factory team.
- AI-driven calculation of DTI, DSCR, and LTV ratios with role-based access control ensuring branch-scoped data governance.

✓ 70% faster document analysis ✓ 75% faster CAM prep ✓ Loan turnaround: 5-7 days → 2-3 days ✓ +40% loan processing capacity

- **Corporate Customer Segmentation Model**

- Delivered end-to-end ML pipeline analysing 59,378 corporate customers across ₦1.88T in deposits, engineering 75+ features from 7.7M transaction records.
- Solved 98.7% class-collapse by designing Stratified Hybrid Clustering — reducing max-to-min cluster ratio from 9,770x to 1.8x.

✓ 4 named actionable segments ✓ ₦48B reactivatable deposits ✓ ₦156B high-value churn protection

- **Enterprise CLV · Channel Recommendation · Behavioural Segmentation · Sentimetrics**

- CLV Model (8.7M customers): Rebuilt predictive CLV generating ₦104.29B turnover uplift, 35,753 customer upgrades to High Value, premium base expanded to 264,000+.
- Channel Recommendation (552,198 customers): Next-best-channel engine generating ₦71.84B transaction value, 73,366 new users onboarded, 231,079 inactive customers converted at 98.1% rate, 5.5M incremental transactions.
- Behavioural Segmentation: Six-category cross-sell framework generating ₦32.89B through 522,818 transactions, 12,100 inactive converted at ₦4.19M average transaction value.
- Sentimetrics: Rebuilt ML NLP architecture, expanded to Facebook, improved classification accuracy for proactive customer dissatisfaction detection.

Grams Technology · Data Scientist *Dec 2020 – May 2022 Ibadan, Nigeria*

- **Grams AI Analytics Agent**

- Designed and deployed a conversational AI analytics assistant democratising internal data access for non-technical staff through natural language SQL querying, RAG-powered document search, and hierarchical RBAC integrated with Azure SQL.
- Built on Azure OpenAI GPT-4, Azure Cognitive Search, and Azure Functions — unifying analytics, knowledge retrieval, and security in one serverless interface, reducing manual data requests and enabling self-service across departments.

GitHub: https://github.com/Cemaster016/Grams-Analytics-AI-Assistant

- **Predictive Analytics & ML Models**

- Designed and implemented advanced ML algorithms achieving 98.5% improvement in predictive accuracy, significantly boosting operational automation across the organisation.
- Elevated big data processing capabilities with 35% accuracy enhancement, enabling data-driven decision-making across key business operations.
- Engineered predictive analytics models for a health initiative accurately forecasting malaria occurrences and optimising antimalarial medication demand, improving healthcare outcomes across target regions.
- Developed and deployed product recommendation models identifying next-best-product opportunities, driving revenue growth and improved customer engagement.

GWS Concept · Machine Learning Engineer *Jun 2022 – Aug 2023 Lagos, Nigeria*

- **Azure-Deployed ML Models & Generative AI**

- Achieved 95% accuracy in new product sales forecasting using advanced predictive analytics, enabling data-driven resource allocation across the organisation.
- Deployed ML models on Azure, automating big data insight extraction and driving a 30% improvement in classification accuracy, streamlining key business processes.

- Spearheaded generative AI integration to forecast electricity demand and supply gaps across African regions, delivering actionable energy management intelligence.
- Deployed product recommendation models identifying next-best-product opportunities, driving substantial revenue increase and improved customer engagement.

SELECTED PROJECTS

Databot — Conversational AI Agent on Power BI — *Wema Bank Plc*
Tech: Azure OpenAI GPT-4o · Azure Functions · Python · Power BI / DAX · Azure Table Storage · Microsoft Fabric

- Enterprise NL2DAX agent enabling 500+ bank staff to query live Power BI dashboards through natural language. 29-rule DAX framework, RLS enforcement, PII-safe schema intelligence, follow-up conversation memory — zero per-dashboard hardcoding across 8+ deployed dashboards.
- Full production stack: serverless HTTP API, schema discovery via COLUMNSTATISTICS(), result caching, audit logging, and conversation memory persisted in Azure Table Storage.

✓ 8+ dashboards live ✓ 500+ users ✓ Sub-5s response

DataPilot — Autonomous ADF Engineering Agent — *Wema Bank Plc*
Tech: Microsoft Copilot Studio · Azure Functions (x15) · Python · OpenAPI/Swagger · asyncio · ADF REST API

- Autonomous agent enabling ADF pipeline creation, monitoring, and failure diagnosis through natural language. Custom OpenAPI connector, SOURCE/SINK separation, async parallel execution, three-tier fuzzy schema mapper.

✓ Pipeline creation: hours → <2 minutes ✓ 15 microservices · 1 connector

MCP Multi-Tool AI Agent — *Personal / Open Source*
Tech: LangGraph · Groq Llama 3.3 70B · Model Context Protocol · FastAPI · SSE · Python

- Production AI agent using MCP for autonomous multi-tool orchestration via natural language. LLM-powered intent routing, 95%+ accuracy, SSE streaming, microservices architecture separating tool execution, orchestration, and UI.

✓ 95%+ routing accuracy ✓ Sub-2s response ✓ Open source
GitHub: https://github.com/Cemaster016/-MCP-Multi-Tool-AI-Agent

Gemini LiveAPI Conversational Voice Engine — *Movables Games (Consulting)*
Tech: Vertex AI · Gemini Live 2.5 Flash · FastAPI · WebSocket · Librosa · Docker · Async Python

- Real-time bidirectional voice API powering AI educational games. Children converse naturally with game characters. Audio normalised to 16-bit PCM at 16kHz. Session-scoped memory. Docker-containerised. Clean WebSocket endpoint abstracting Gemini LiveAPI complexity for game developers.

Azure AI Educational API Service — *Movables Games (Consulting)*
Tech: Azure AI Foundry · GPT-4o · DALL-E 3 · Flask · Azure AI Inference SDK

- REST API exposing Chat Completion and Text-to-Image endpoints powering in-game tutors, hint systems, and real-time visual asset generation. Eliminates static content dependency, enabling personalised AI-adaptive education at scale.

Grams AI Analytics Agent — *Grams Technology*

Tech: Azure OpenAI GPT-4 · Azure SQL · Cognitive Search · Azure Functions · Python

- Conversational analytics assistant enabling NL-to-SQL data queries, RAG-powered document retrieval, and hierarchical RBAC — unifying analytics, knowledge, and security in one serverless interface.

GitHub: https://github.com/Cemaster016/Grams-Analytics-AI-Assistant

Personal Loan Prediction Model — *AllLife Bank*
Tech: scikit-learn · Decision Trees · Flask · Python

- Decision tree classifier: 94.8% test accuracy, 98.6% recall. Deployed as Flask API enabling bank staff to precisely target high-probability loan acceptance customers.

GitHub: https://github.com/Cemaster016/Personal-Loan-Prediction-Pro/tree/main

EDUCATION

MSc Data Science & Technologies (In Progress) *2025 – Present*
University of East London, UK · Research: AI applications in Special Educational Needs & Disabilities (SEND)

B.Tech Industrial Mathematics *2014 – 2019*
Federal University of Technology Akure (FUTA), Nigeria

CERTIFICATIONS & TRAINING

- **Building AI Agents with Azure AI Studio** — Microsoft
- **Advanced LangChain & Prompt Engineering** — HamoyeAI
- **AWS Machine Learning Foundational** — Udacity / Amazon
- **Advanced Data Science** — Bluechip Technologies
- **Azure Fundamentals** — Microsoft / LinkedIn Learning
- **Advanced Data Engineering** — Bluechip Technologies
- **Reporting & Analytics with Power BI** — DBrown Consulting
- **Data Science Professional** — Delon Group

AWARDS

Best in Data Science / AI — *Wema Bank Plc (March 2024)*
Awarded for outstanding resilience, hard work, and benchmark-setting performance in AI and data science across the intensive Data Science & AI training programme.

PORTFOLIO & LINKS

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